

JS Exam System – Lab

1. System Overview

The JS Exam System is a browser-based interactive exam application built using ES6+ JavaScript classes and modules.

It allows users to answer questions sequentially, navigate between them, and view their final score at the end of the exam.

The system supports:

- True/False questions
- Short answer questions
- Score calculation based on grades

2. Functional Requirements

2.1 Question Management

- The system shall allow creating questions using a Question class.
- Each question must contain: id, text, correctAnswer, grade.
- The class must provide a method to check answers.
- Answer comparison must be case-insensitive.

2.2 Exam Management

- The system shall use an Exam class to manage the exam.
- The Exam class must store a list of questions and student answers.
- It must provide functionality to add questions.

2.3 Answer Handling

- The system shall allow the user to input answers for each question.

- Answers must be automatically saved when navigating between questions.
- Users must be able to review and modify answers before finishing the exam.

2.4 Navigation

- The system shall support Next and Previous buttons.
- Next moves to the next question.
- Previous moves to the previous question.
- On the last question, clicking Next shows the results.

2.5 Result Calculation

- After completing the exam, the system compares answers with correct answers.
- The system calculates the total score.
- Results per question must be displayed with correct or incorrect indicators.

2.6 Restart Function

- The system shall provide a Restart button that clears answers and restarts the exam.

3. User Interface Requirements

- The interface must display one question at a time.
- The system must show a progress indicator such as 'Question 1 of 5'.
- The design should include:
 - Gradient background
 - Card-based layout
 - Responsive design

4. System Architecture

The system follows Object-Oriented Programming (OOP) principles.

Components:

- Question Class
- Exam Class
- UI Layer (HTML)
- Application Logic (JavaScript)

5. Module Structure

The project uses ES Modules.

File structure:

- index.html (UI layout)
- app.js (Application logic and DOM manipulation)
- exam.js (Exam class definition)
- question.js (Question class definition)
- TODO.md (Progress tracking)

6. Sample Exam Questions

Total grade: 45 points

1. Is JavaScript fun? → true (5)
2. $2 + 2 ? \rightarrow 4$ (10)
3. What is ES6? → ecmascript 2015 (10)
4. Classes are sugar? → true (5)
5. Modules keyword? → import (15)